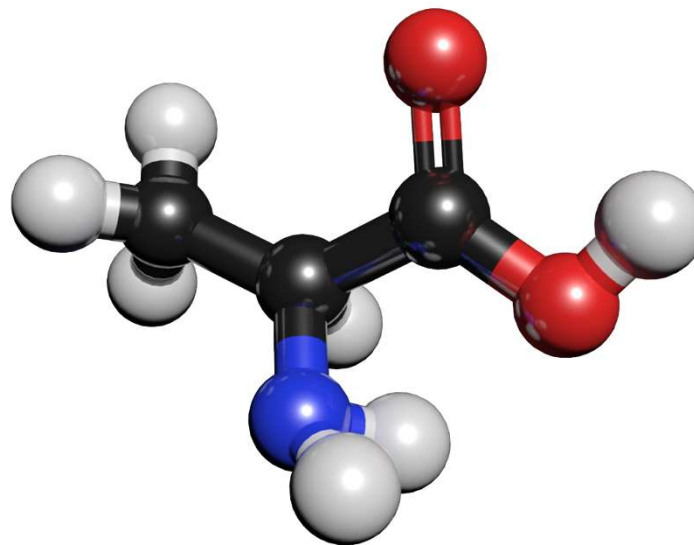
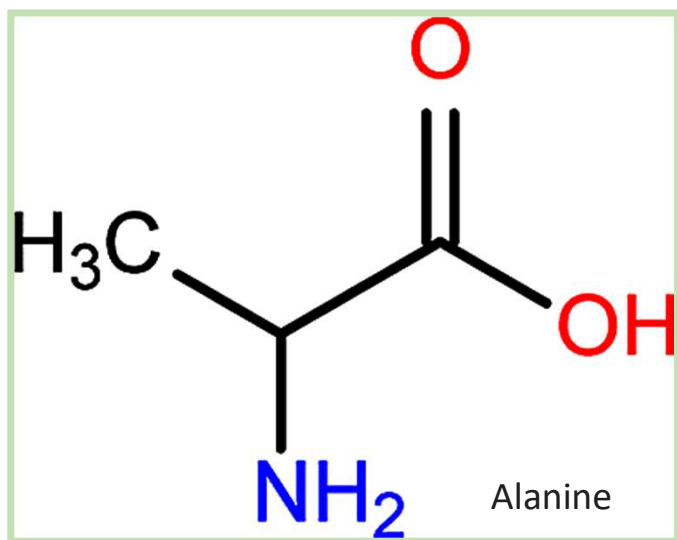
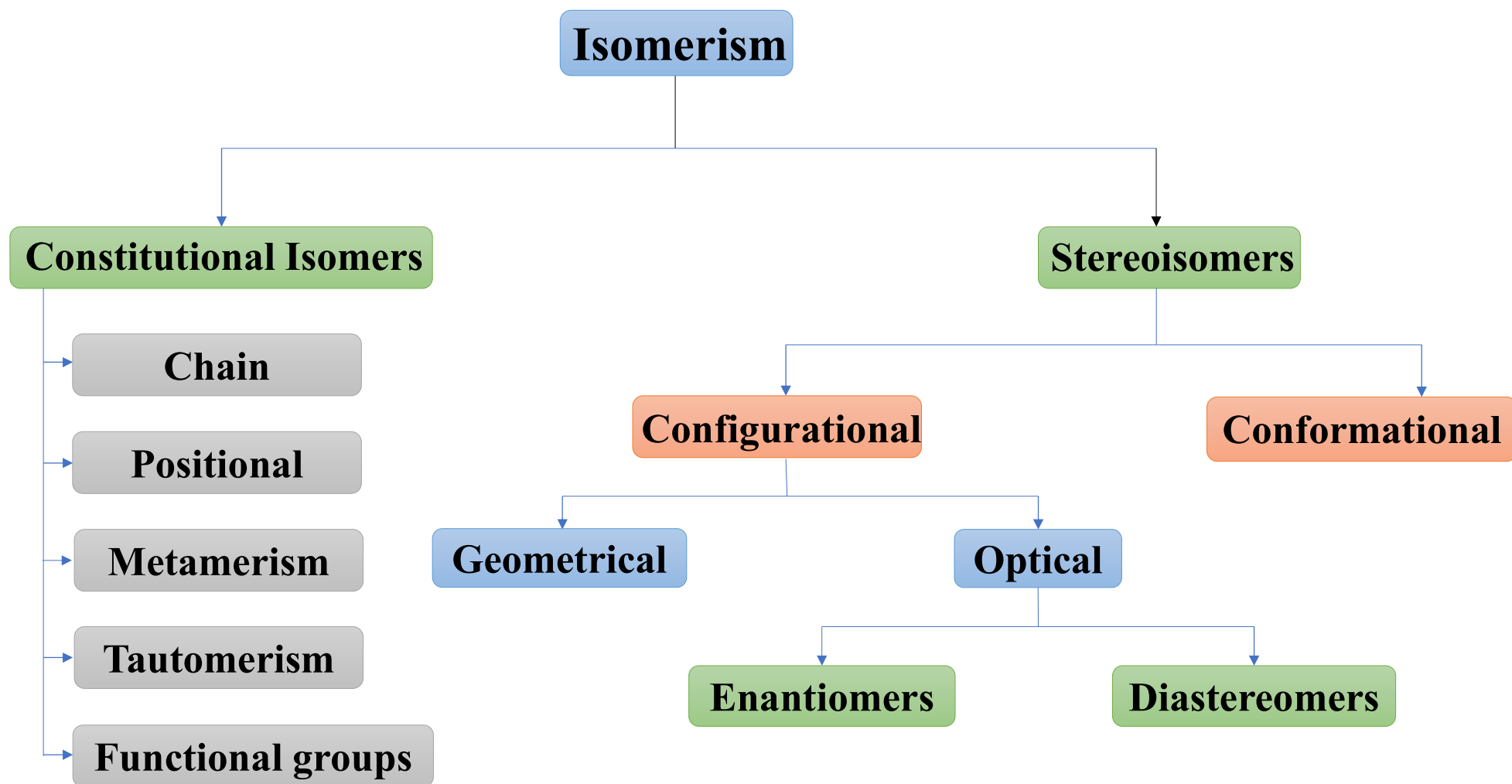


Stereochemistry

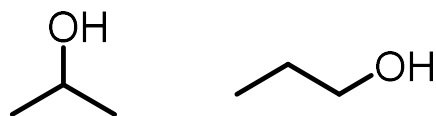
It's the study of three-dimensional structure of molecules.





Constitutional : isomers differ in the order in which the atoms are connected so they can contain different functional groups and/ or bonding patterns (*e.g.* branching)

example: 1-propanol, 2-propanol

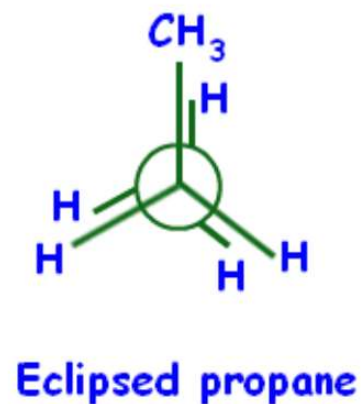
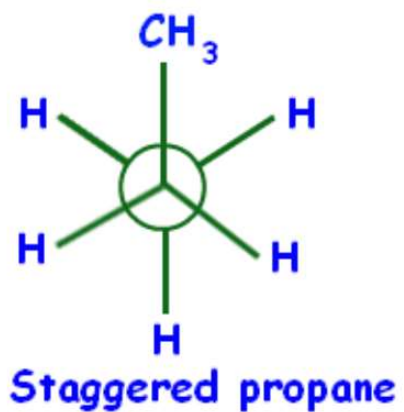


Stereoisomers: have the same functional groups and the same atoms order, they differ only in the arrangement of atoms and bonds in space.

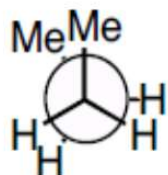


Conformational isomers

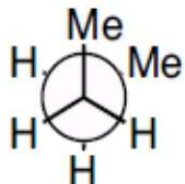
- Stereoisomers that are produced by rotation about bond (single bond), and are often rapidly interconverted at room temperature.



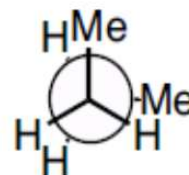
Butane conformational isomers



fully eclipsed
(eclipsed)
 $\theta_{\text{Me/Me}} = 0^\circ$



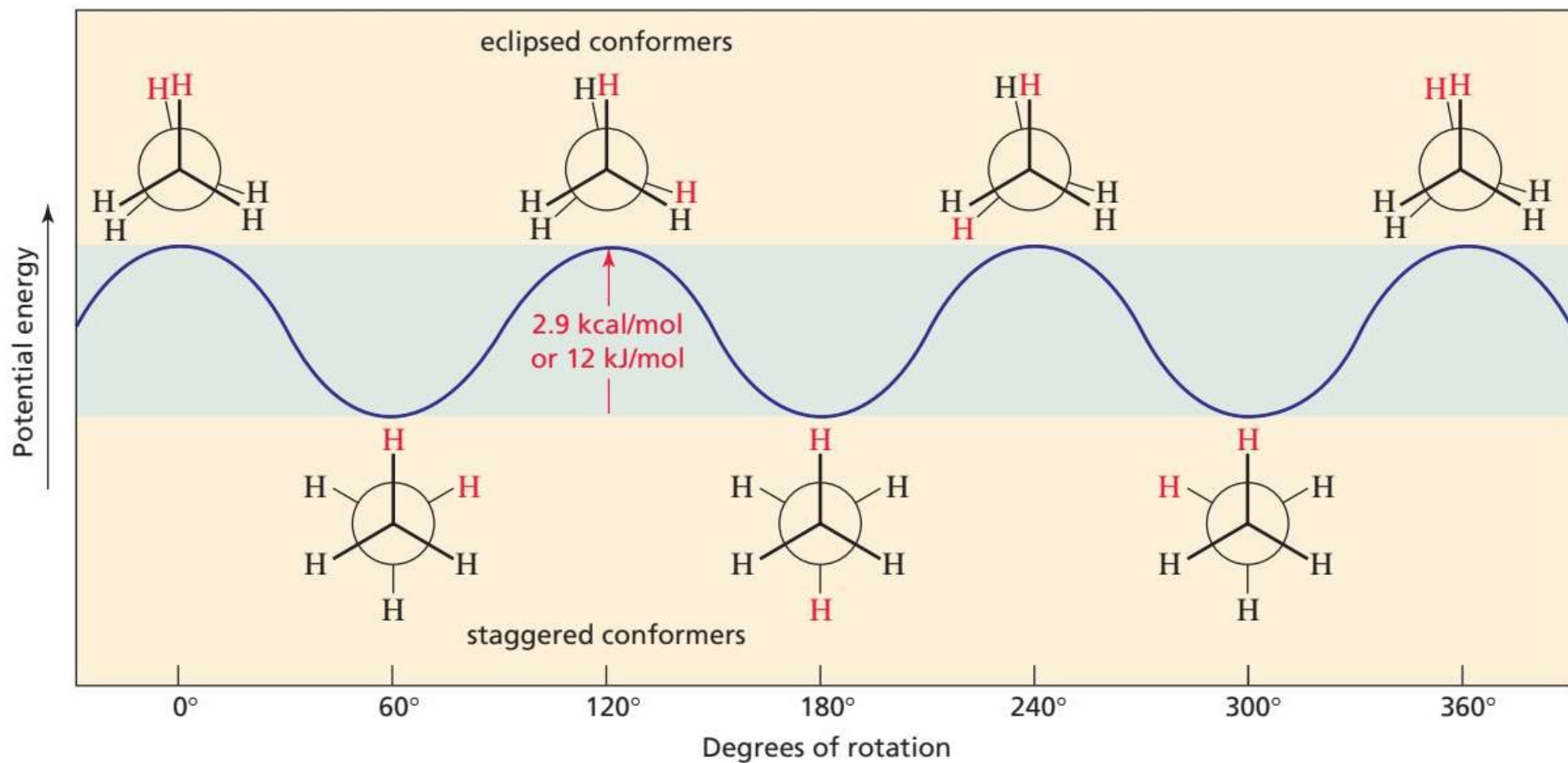
gauche
(staggered)
 $\theta_{\text{Me/Me}} = 60^\circ$



eclipsed
(eclipsed)
 $\theta_{\text{Me/Me}} = 120^\circ$



anti
(staggered)
 $\theta_{\text{Me/Me}} = 180^\circ$



▲ **Figure 2.4**

Potential energy of ethane as a function of the angle of rotation about the carbon-carbon bond.

Configurational isomers

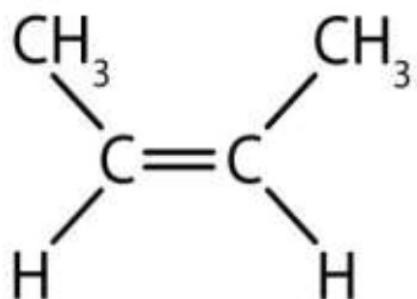
Stereoisomers that do not readily interconvert at room temperature and can be separated.

Geometric isomers

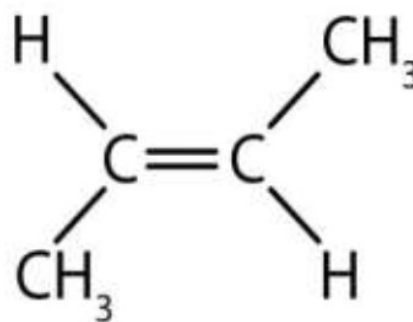
Optical isomers

Geometric isomers

These isomers occur where you have restricted rotation somewhere in a molecule (ex double bond) or across a ring system



cis-2-butene



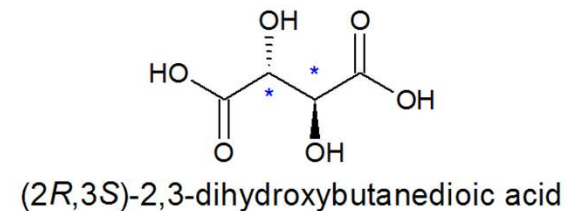
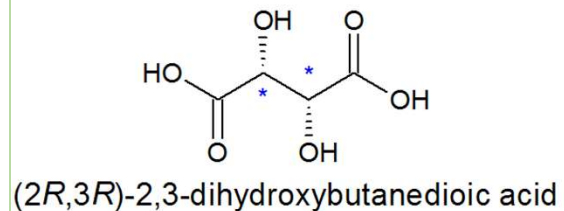
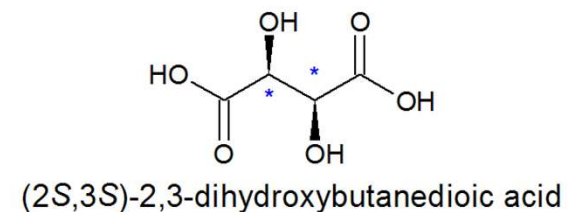
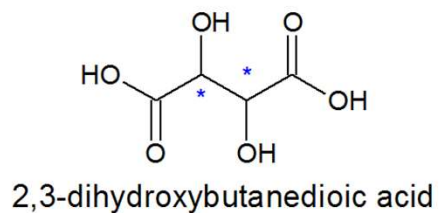
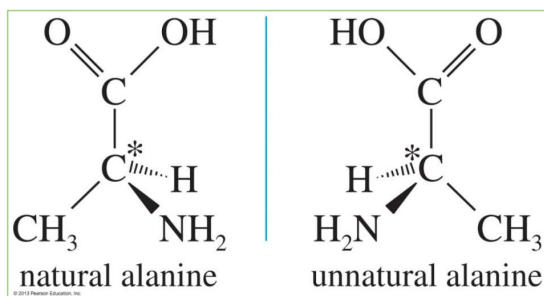
trans-2-butene

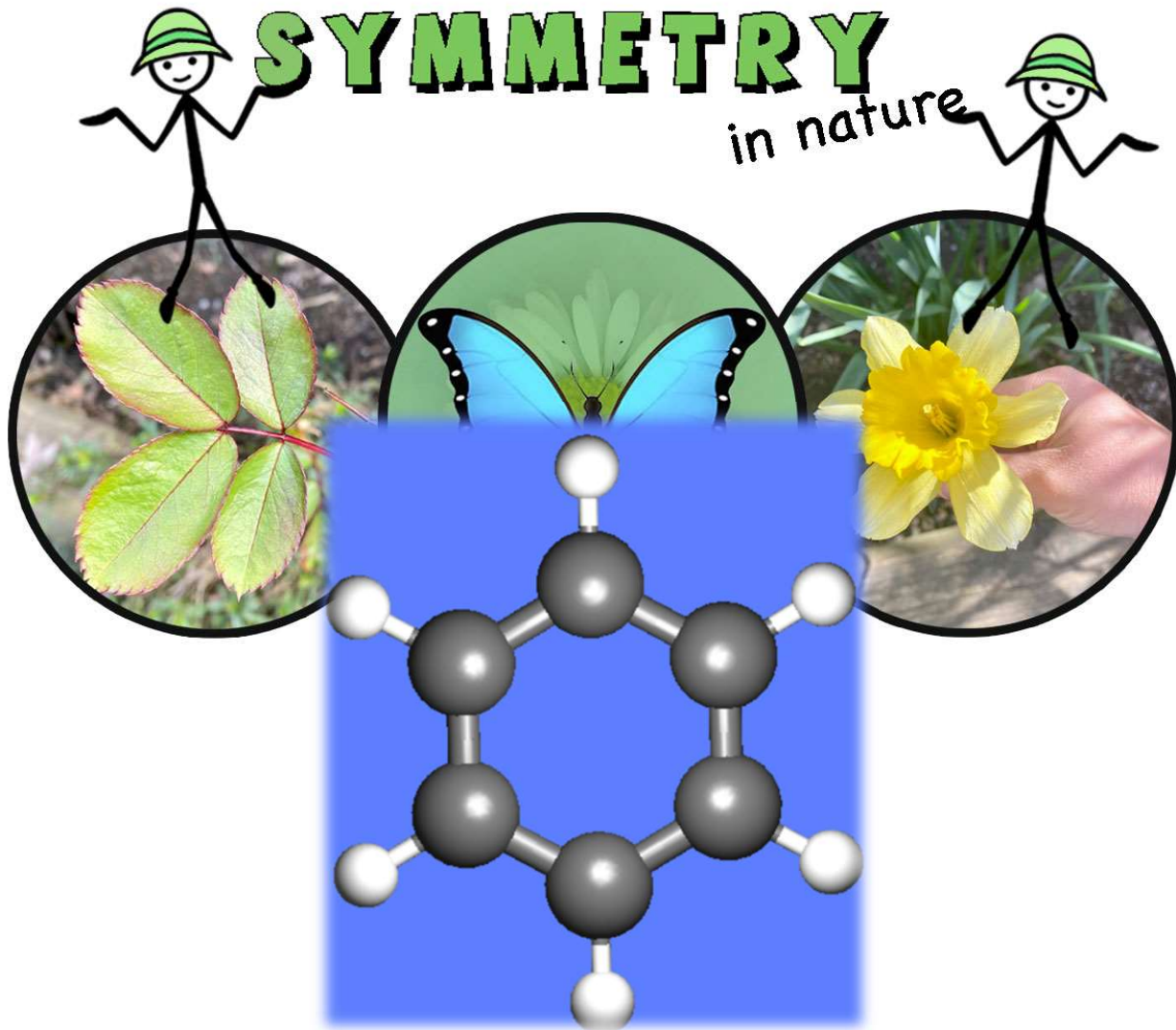
Optical isomers

Configurational isomers that differ in the 3D relationship of the substituent about one or more atoms.

Enantiomers

Diastereomers



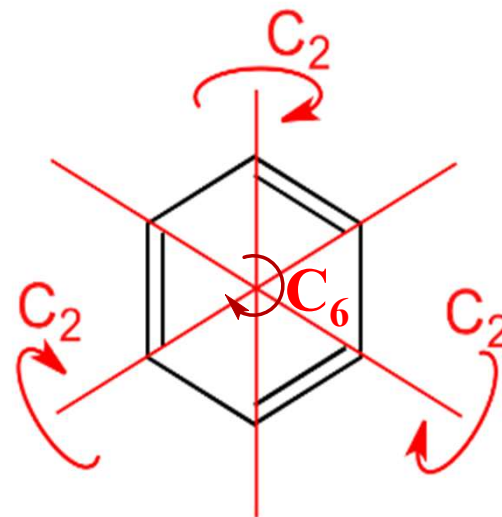
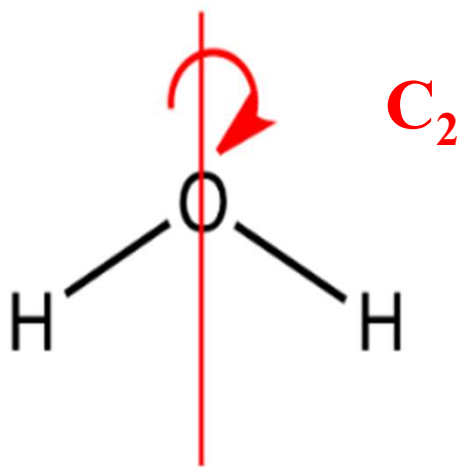
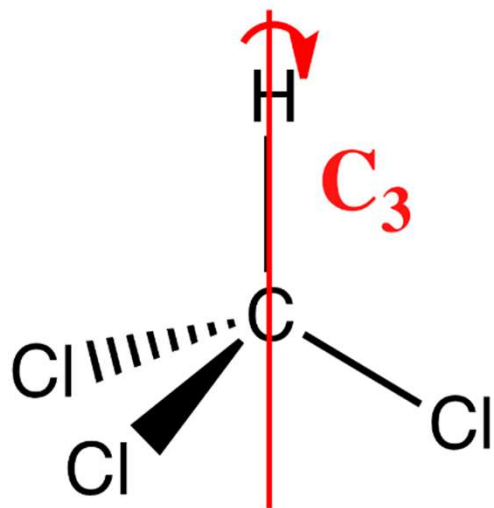


Symmetry

- **Symmetry element** is a geometrical entity such as a **plane, a line, a point** with respect to which after operation we get **superimposable on the original one**.
- There are four types of symmetry elements. These are
 - (a) **Simple or proper axis of symmetry (C_n)**
 - (b) **Plane of symmetry (σ)**
 - (c) **Centre of symmetry (i)**
 - (d) **Alternative axis of symmetry (S_n)**

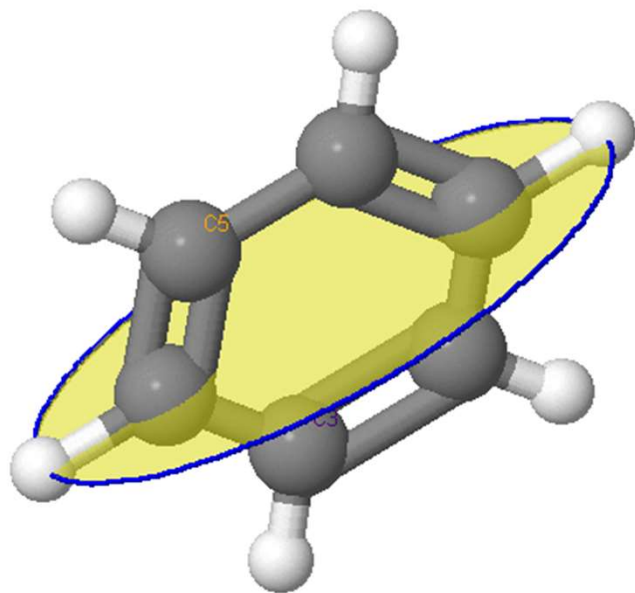
□ Simple or proper axis of symmetry (C_n)

If the molecule is **rotate by an angle of $360^\circ/n$ around an imaginary line (axis)** passing through the molecule, an indistinguishable structure from original results.

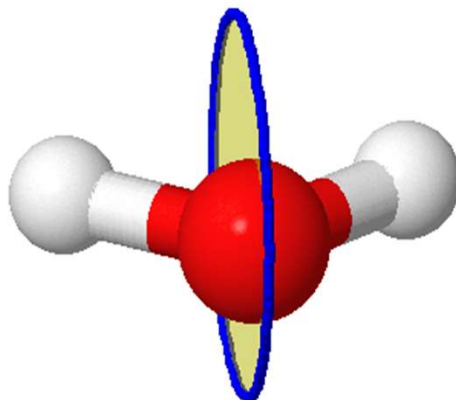


□ Plane of symmetry (σ)

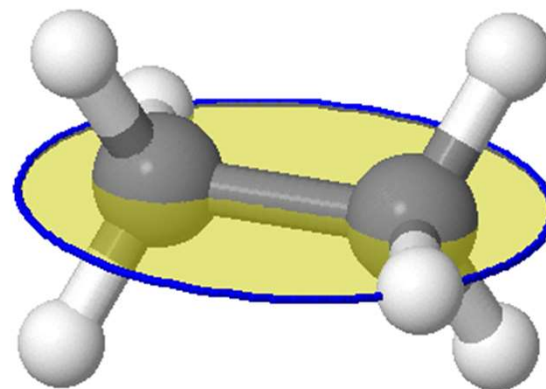
An imaginary plane that **bisects a molecule** in such a way that one half of molecule is mirror image of other half (**plane acting as mirror**)



Benzene



Water



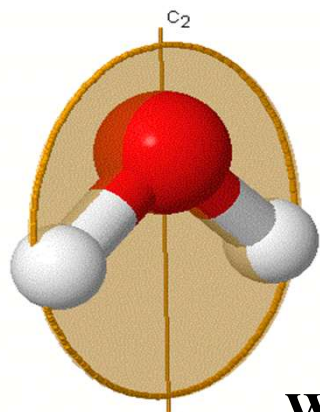
Ethane

Classification of plane of symmetry

Depending upon **the position of σ**
with respect to **principal axis**

σ_v

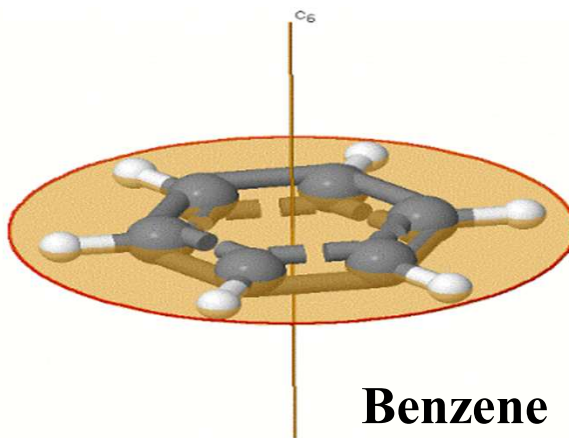
➤ Plane **containing**
principal axis



Water

σ_h

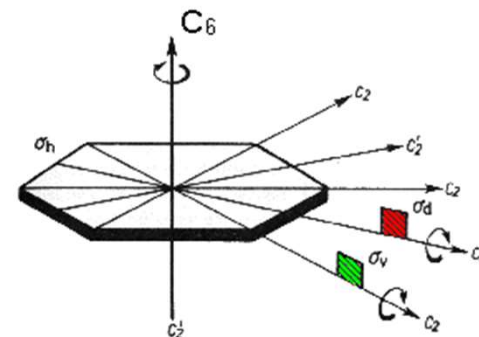
➤ Plane **perpendicular**
to principal axis



Benzene

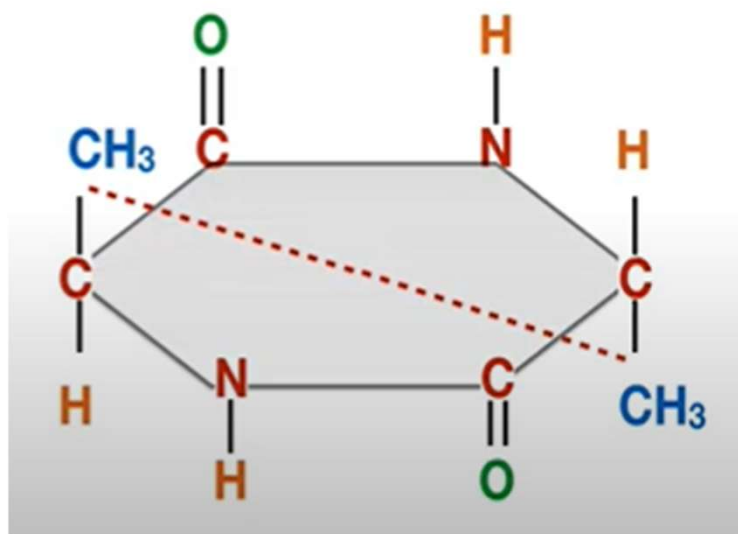
σ_d

➤ Plane which **contains**
principal axis **and bisects**
angle between two C₂



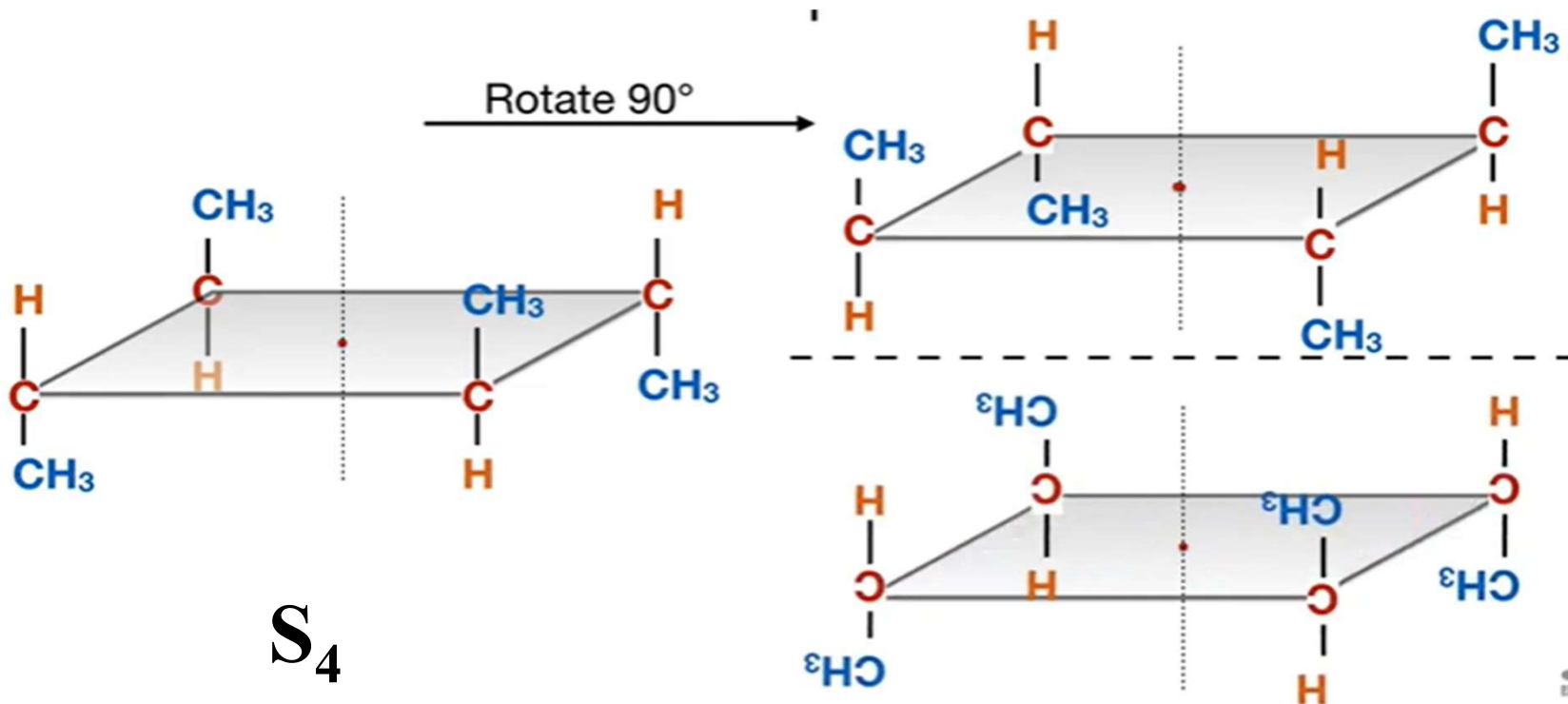
□ Centre of symmetry (i)

A point within a molecule such that if a **straight line is drawn from any atom of the molecule to the point and then extended equal distance beyond the point**, another identical atom will be found at the end of the line.

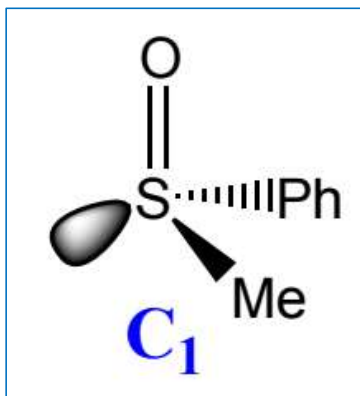


□ Alternative axis of symmetry (S_n)

An imaginary axis in molecule such that **rotation of the molecule about the axis by $360^\circ/n$ followed by reflection in a plane perpendicular to this axis** generates a indistinguishable structure to original.

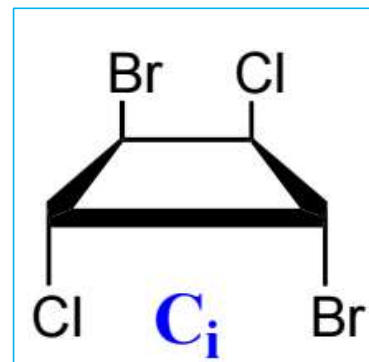


Symmetry elements and Point Group:



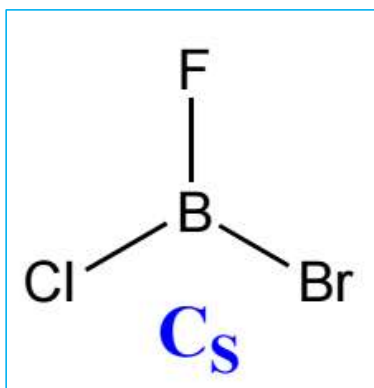
Sym elements

E

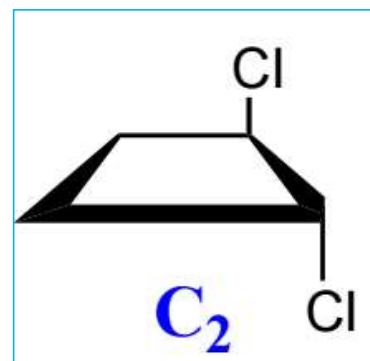


Sym elements

i



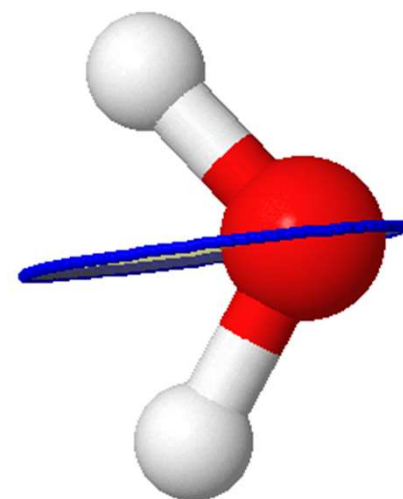
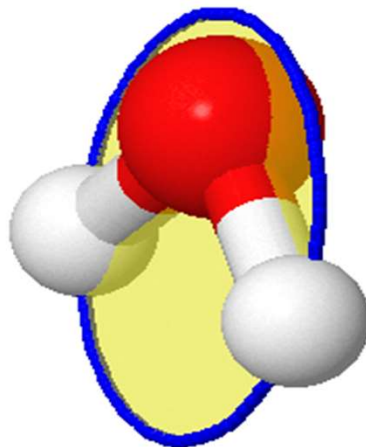
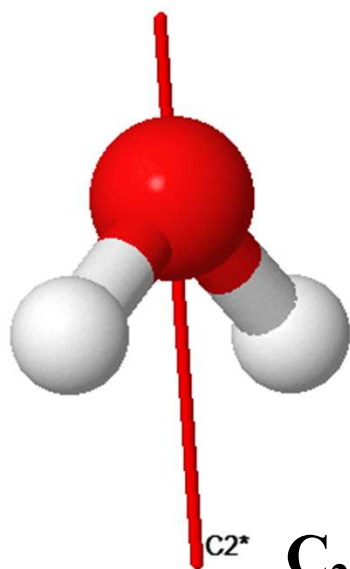
σ



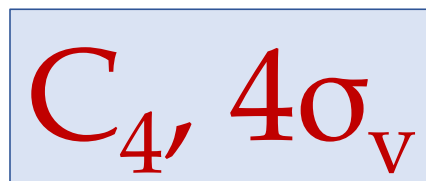
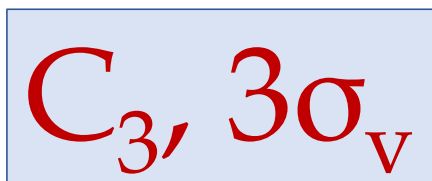
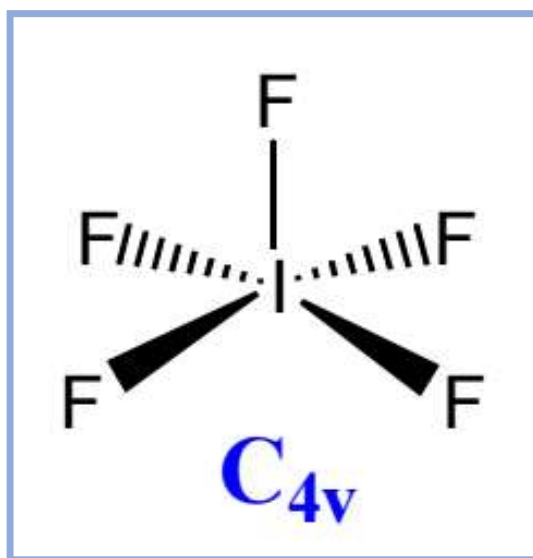
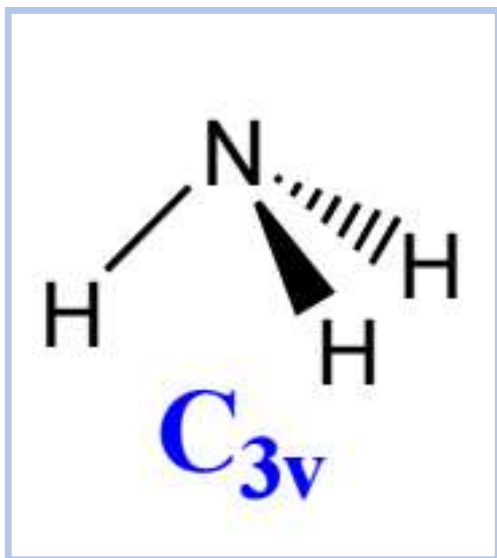
C_n

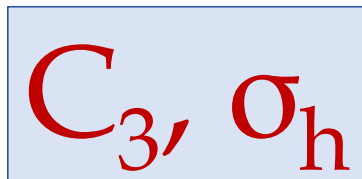
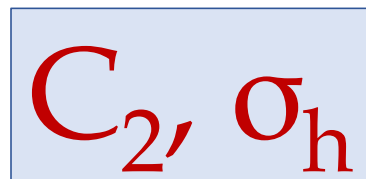
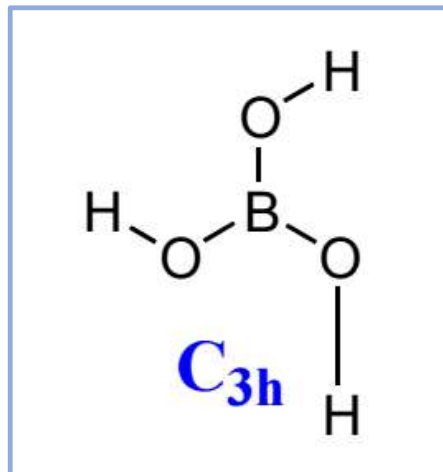
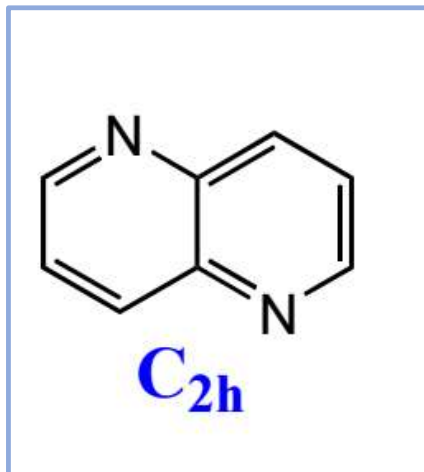
Point group of H₂O

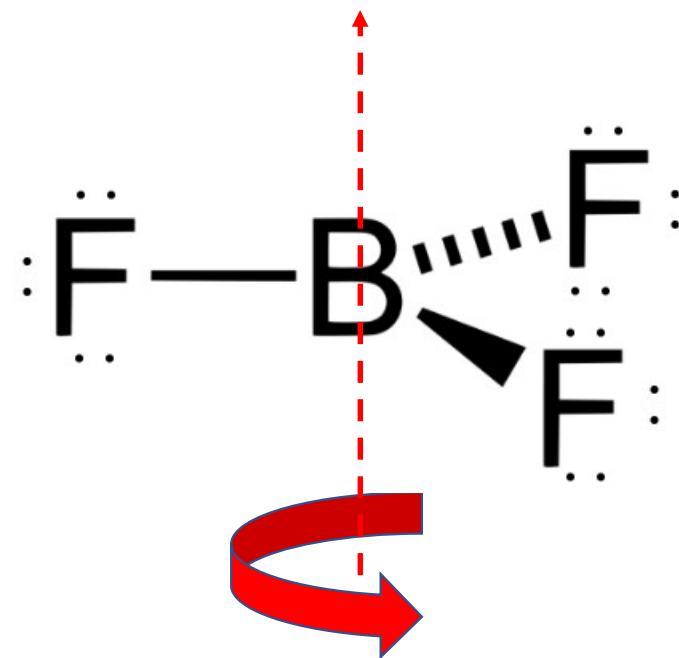
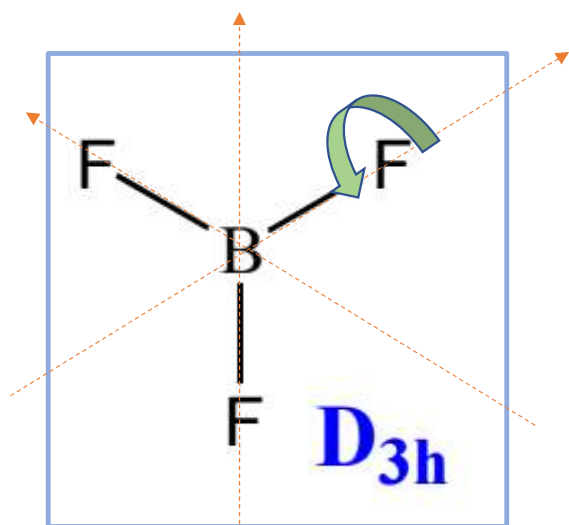
➤ Symmetry elements :



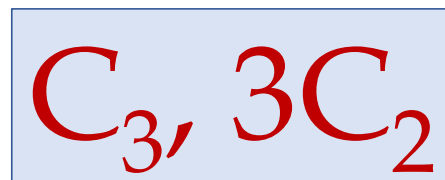
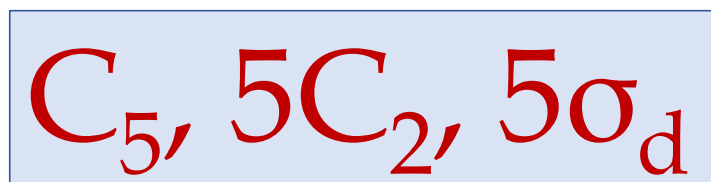
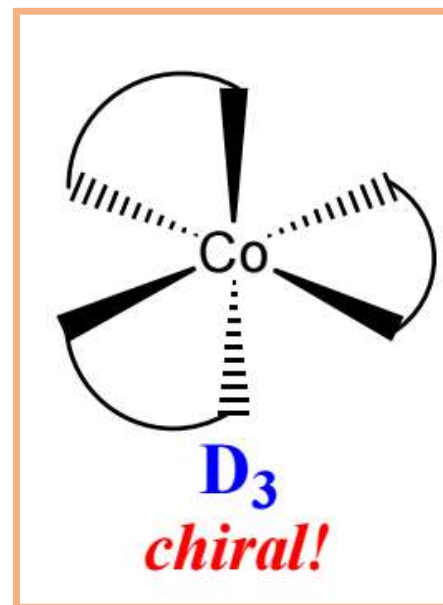
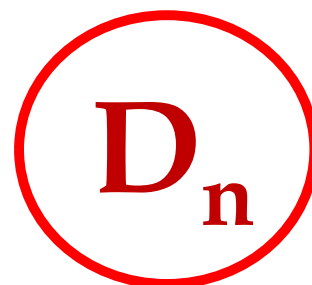
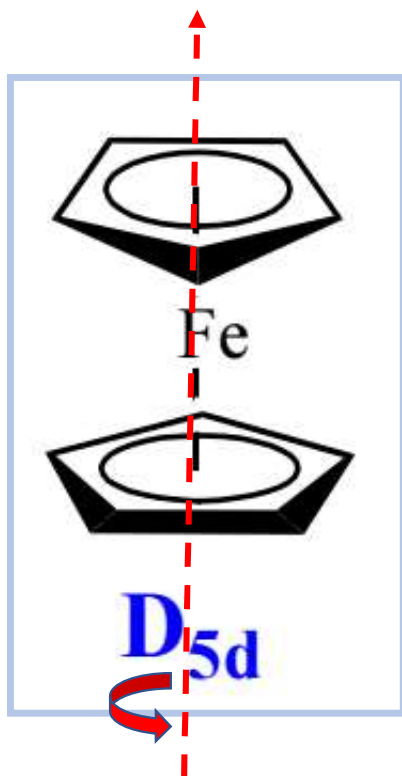
Point group is C_{2v} (C_{nv})





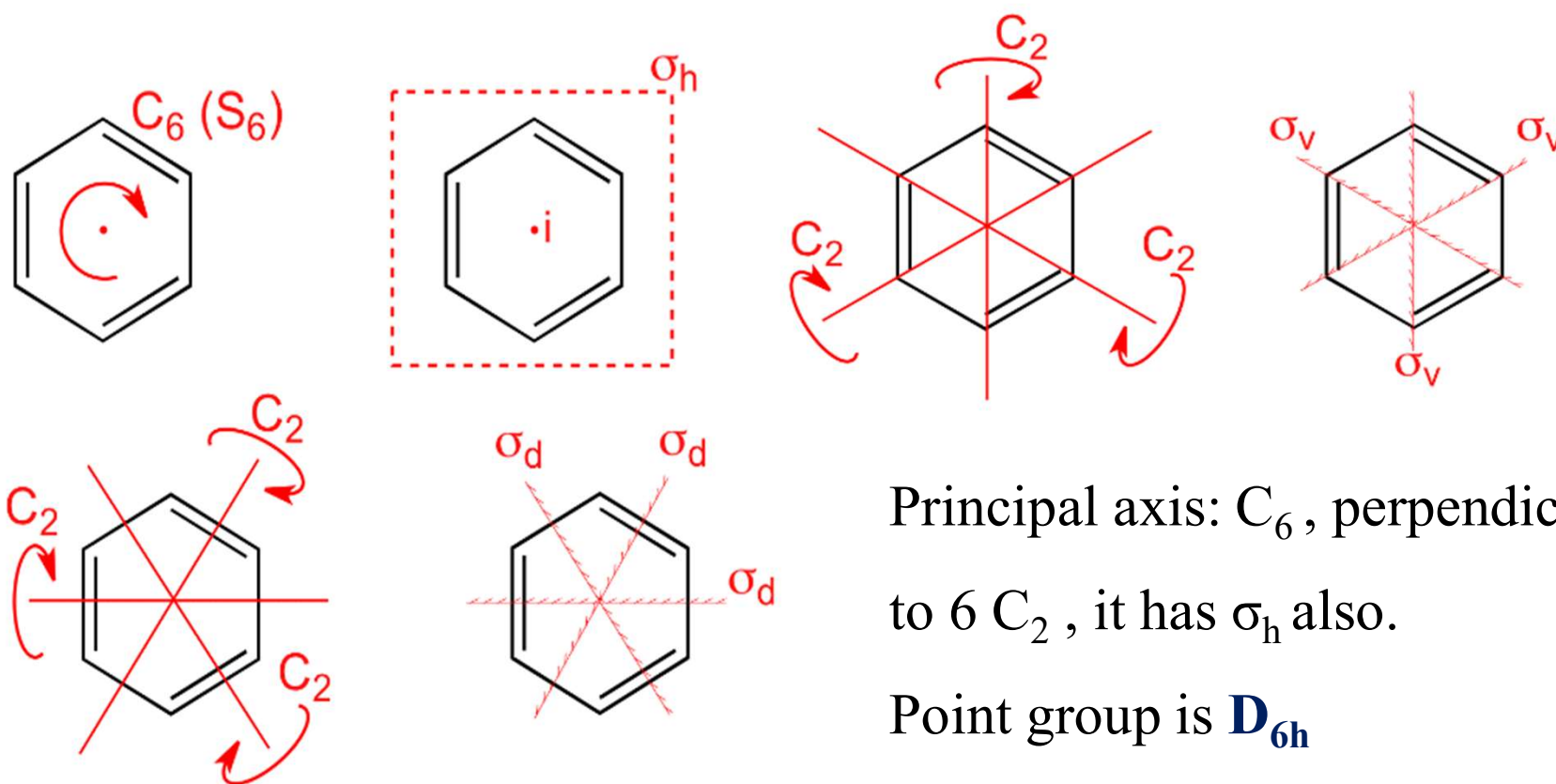


$C_3, 3C_2, 3\sigma_v$ and σ_h



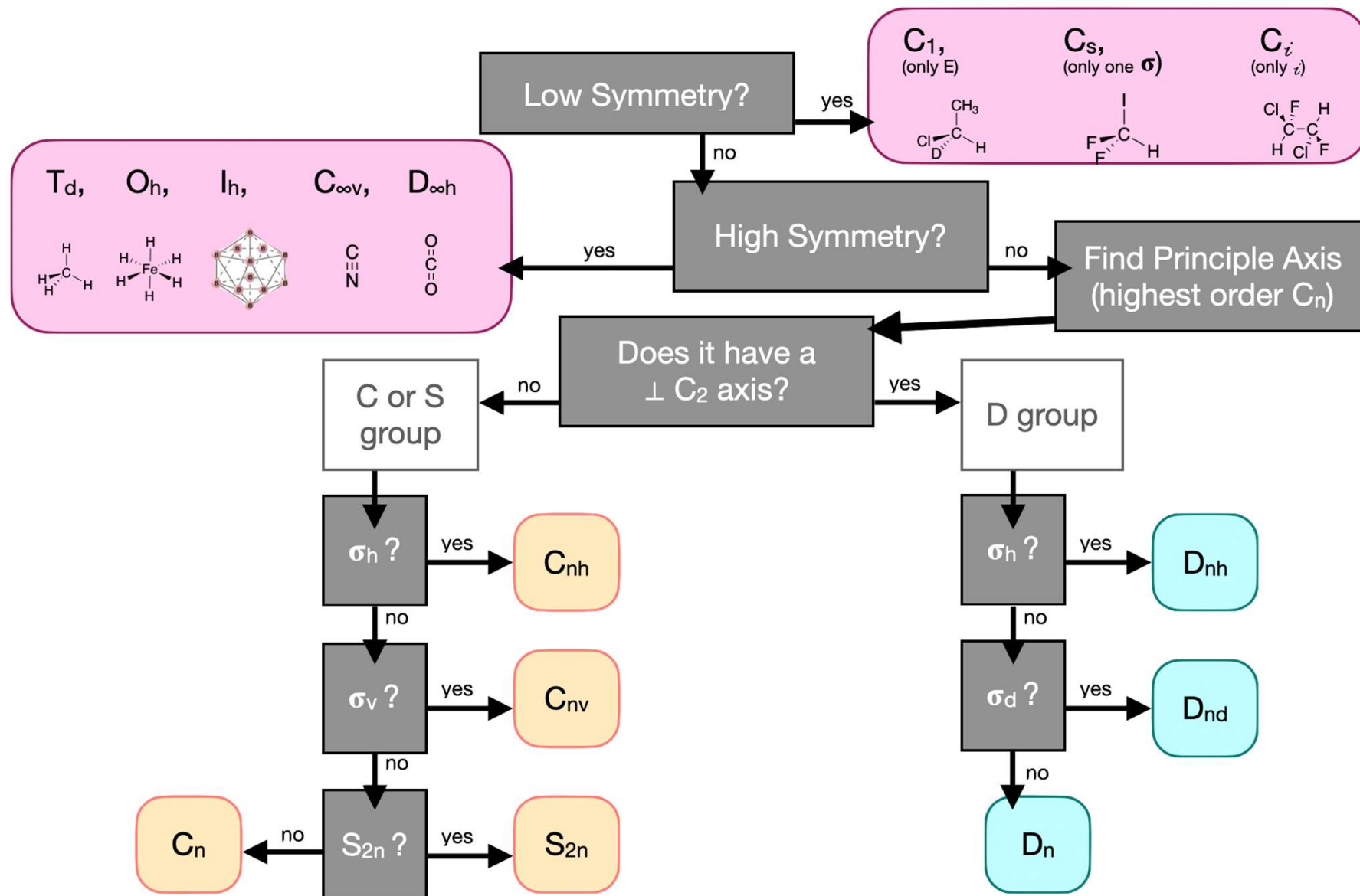
Point group of benzene (C_6H_6)

➤ Symmetry elements:



Principal axis: C_6 , perpendicular to 6 C_2 , it has σ_h also.

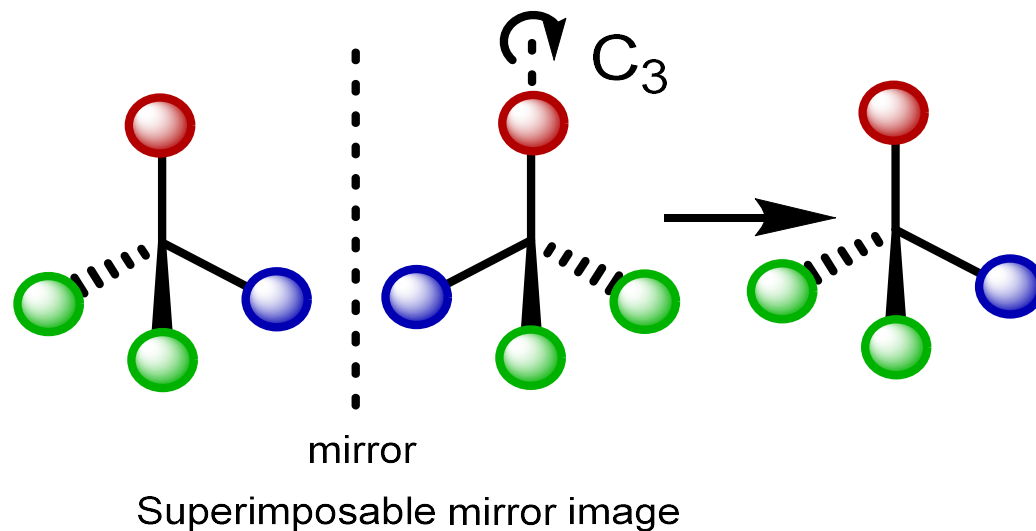
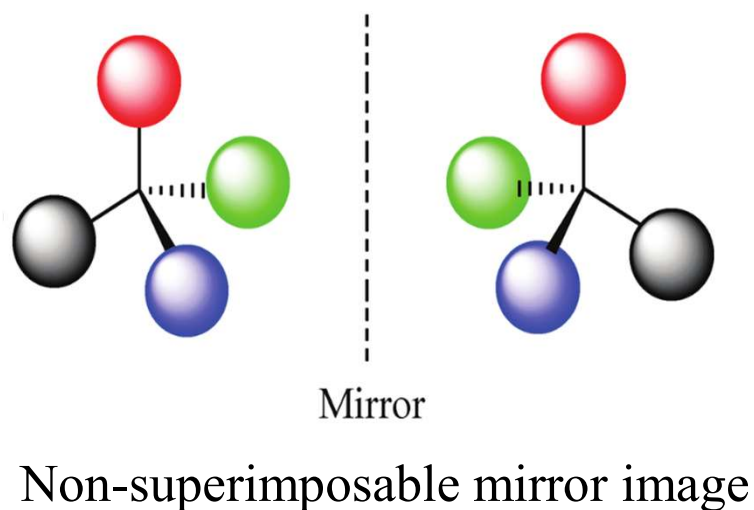
Point group is **D_{6h}**



Point gr.	Sym elements*	h***	Point gr.	Sym elements*	h***
C_1	E	1	C_i	i	2
C_s	σ	2	C_n	C_n	n
S_n	S_n	n	C_{nv}	$C_n, n\sigma_v$	2n
C_{nh}	C_n, σ_h	2n	D_n	$C_n, nC_2 \perp C_n$	
D_{nd}	$C_n, nC_2 \perp C_n, n\sigma_d$	4n	D_{nh}	$C_n, nC_2 \perp C_n, \sigma_h, n\sigma_v$	4n
$C_{\infty v}$	linear no i	∞	$D_{\infty h}$	linear with i	∞

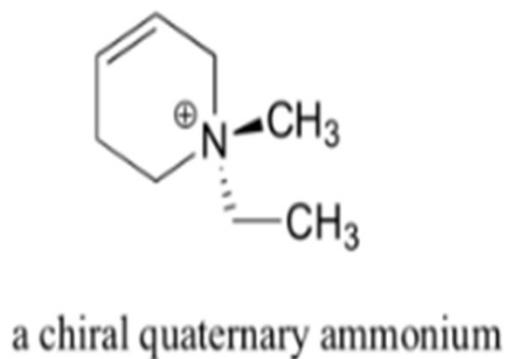
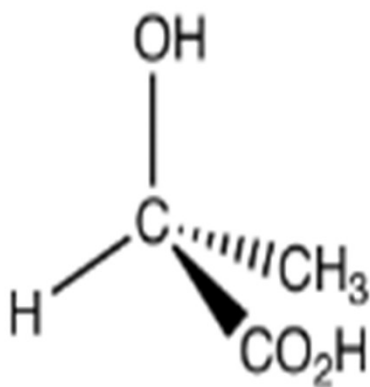
Chiral molecule:

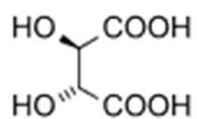
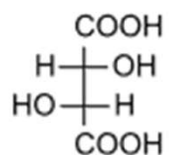
- If the **mirror image** of a molecule is **not superimposable on the original**.
- If the molecule **doesn't have** σ plane of symmetry, inversion centre, Alternative axis of symmetry, then it is said to be chiral molecule.
- Chiral molecule **may have** C_n ($n > 1$) **axis** of symmetry.
- Chiral molecules **rotate the plane** of plane polarized light.



Chiral centre:

- Chiral centre is an atom **bonded tetrahedrally to four different atoms or groups**. Chiral centre may be **C, N, P, S, Si** etc.
- If the molecule has **one chiral centre** it **must be** chiral molecule. But if **multiple chiral centres** are present, then it **may not be** chiral molecule.

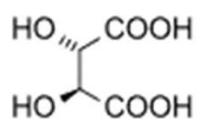
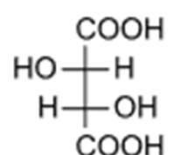




(natural) tartaric acid

L-(+)-tartaric acid

dextrotartaric acid



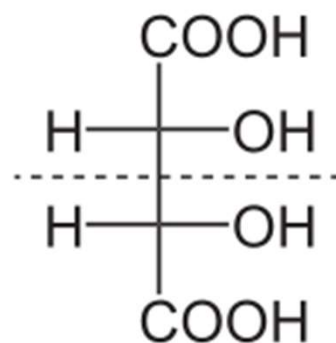
D-(-)-tartaric acid

levotartaric acid

(1:1)

DL-tartaric acid

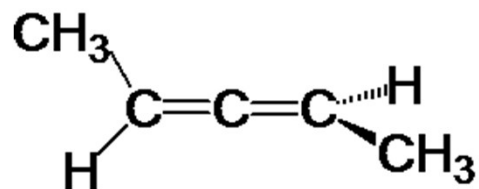
"racemic acid"



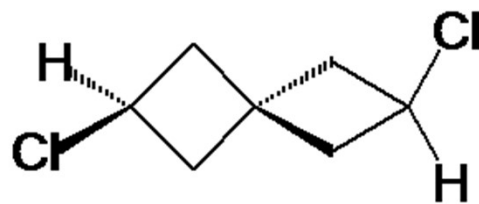
mesotartaric acid

Chiral molecule having no chiral centre:

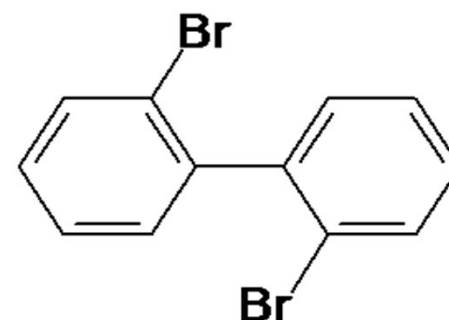
➤ Axial chirality:



Dimethylallene

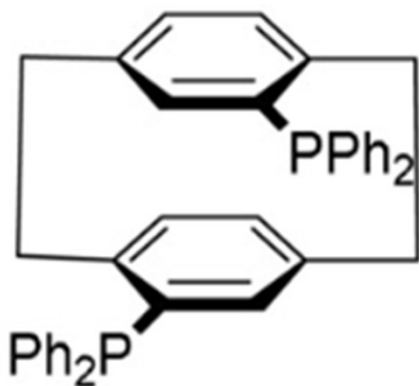


Dichlorospiroheptane

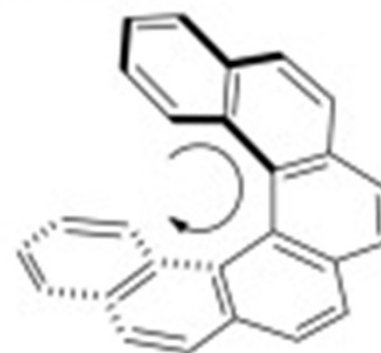
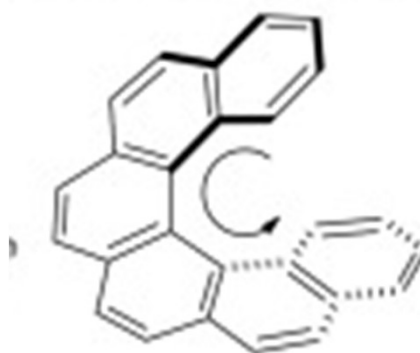


Dibromobiphenyl

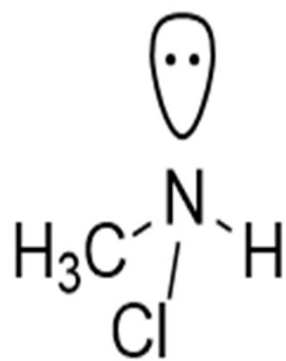
➤ Planner chirality:



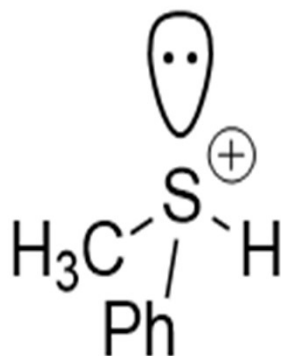
➤ Helical Chirality:



Condition of the molecule to be chiral including lone pair

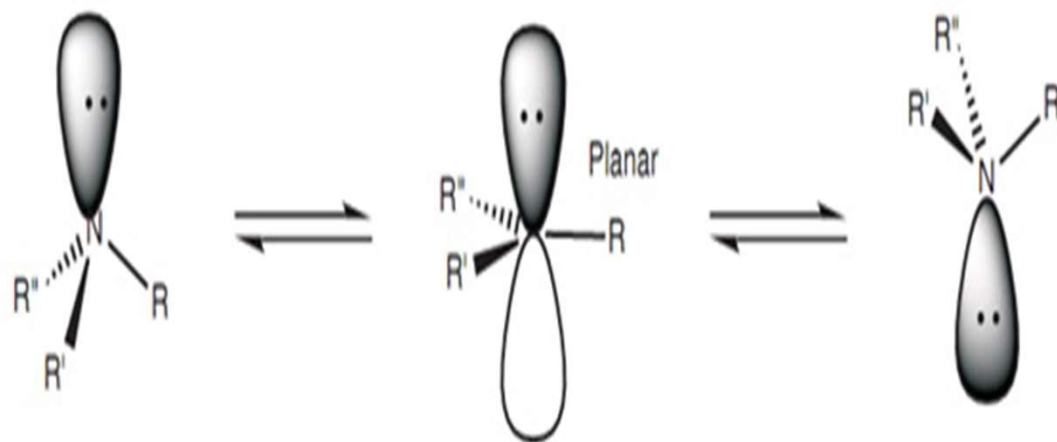


Achiral



chiral

Pyramidal inversion or umbrella inversion



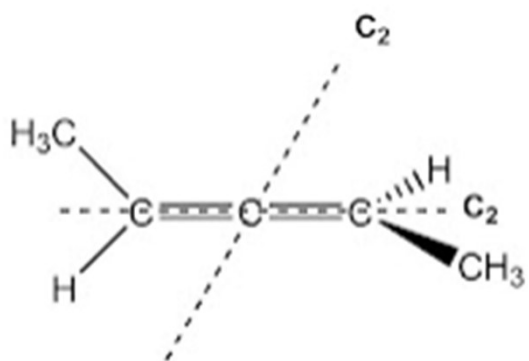
Rapid interconversion at room temperature forms racemic mixture.

Stereogenic centre or Stereocentre:

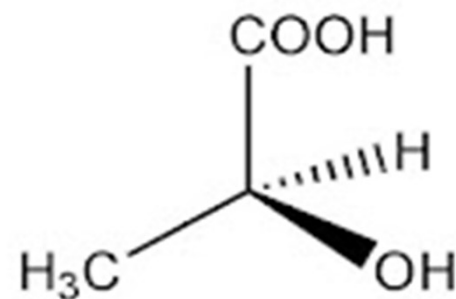
- If any two ligands attached to an atom are interchanged spatially resulting a new stereoisomer.

Asymmetric and Dissymmetric molecules:

- molecule without any C_n axis ($n > 1$) of symmetry, called **Asymmetric molecule**
- molecule with C_n axis ($n > 1$) of symmetry, called **dissymmetric molecule**.
- **All are chiral molecules.**



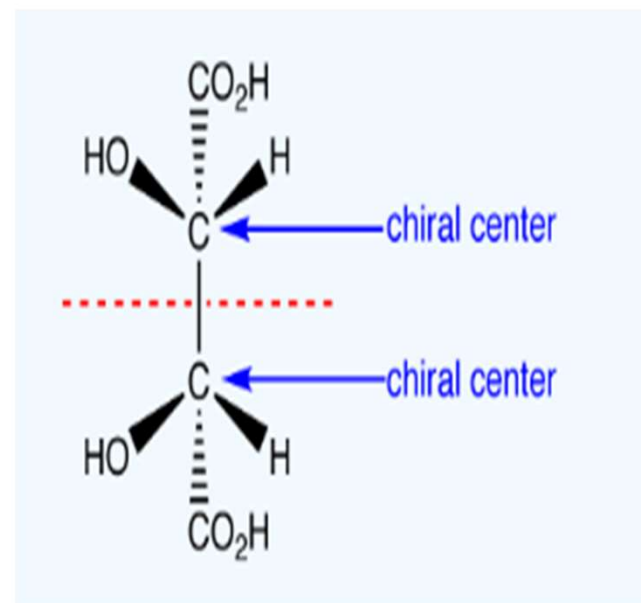
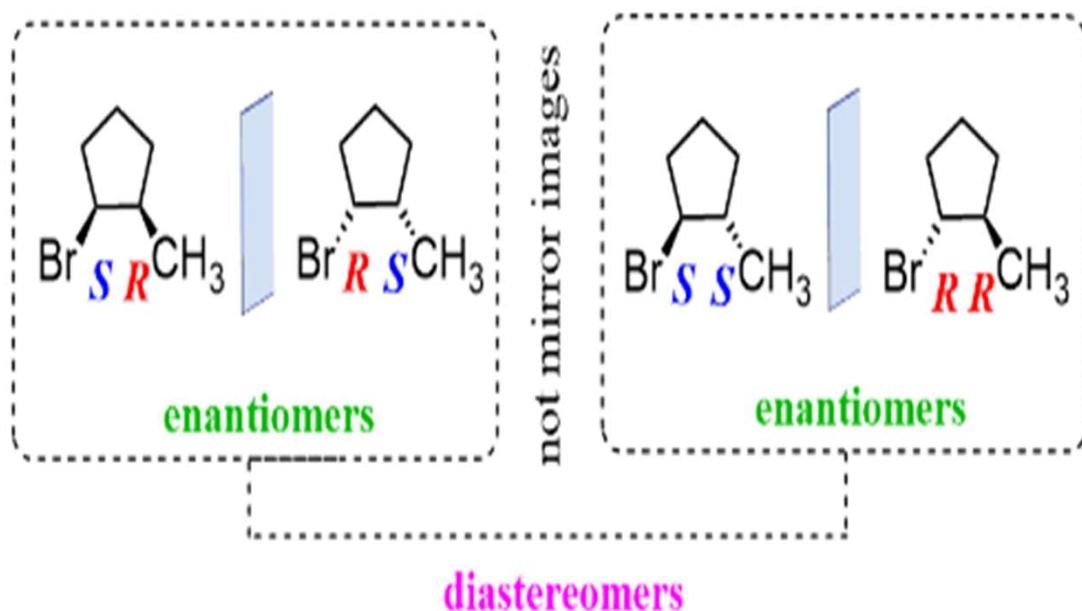
Dissymmetric molecule



Asymmetric molecule

Enantiomers, Diastereomers and Meso isomers

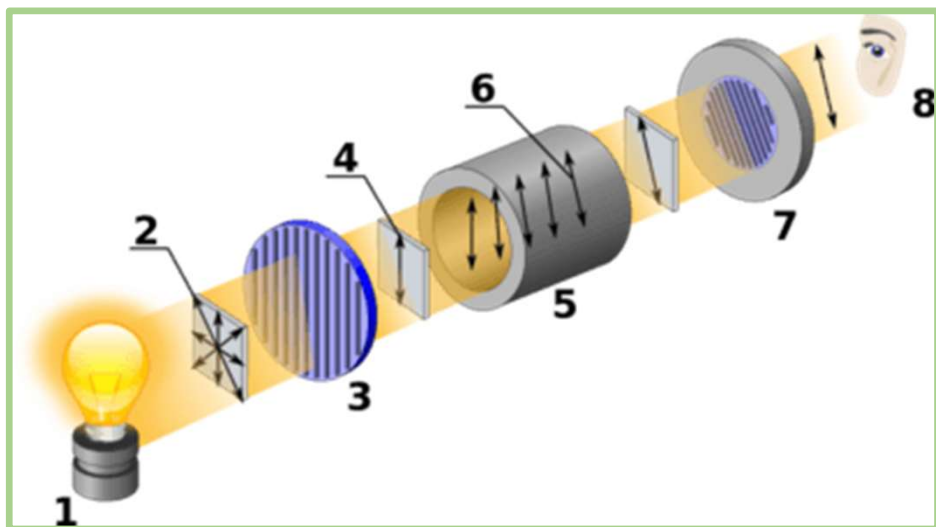
- ❑ Enantiomers: Isomers having non-superimposable mirror image relationship.
- ❑ Diastereomers: Isomers having no mirror image relationship.
- ❑ Meso Isomers: Molecules having multiple stereocentres and a plane of symmetry.



Meso Isomer

Optical Activity

Optical activity is the ability of a chiral molecule to rotate the plane of plane-polarised light



Formula for Specific Rotation

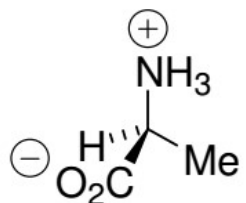
$$[\alpha] = \frac{\alpha_{\text{observed}}}{c \times l}$$

Specific rotation (in degrees*)

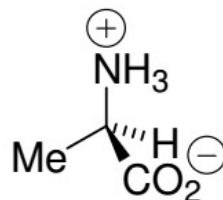
Observed rotation (units = degrees)

Concentration (units = g/ml)

Path length (units = dm)



(+)-alanine $[\alpha]_D = + 8.5^\circ$



(-)-alanine $[\alpha]_D = - 8.5^\circ$

Racemic
50:50

If one enantiomer is in excess?

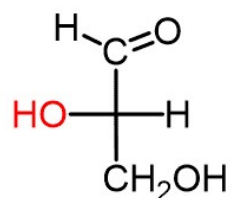
Enantiomeric Excess (ee)

$$\% \text{ ee} = 100 (\alpha_{\text{observed}} / \alpha_{\text{specific}})$$

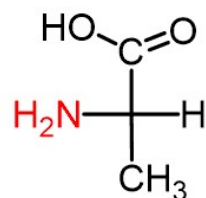
$$\begin{aligned} \% \text{ ee} &= \% \text{ of R} - \% \text{ of S} \\ &= 100 \times (R - S) / (R + S) \end{aligned}$$

R/S Nomenclature according to CIP rule

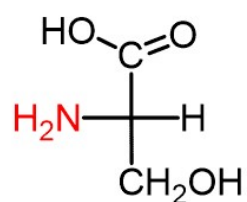
➤ Developed by three chemists, R.S. Cahn, C. Ingold, V. Prelog



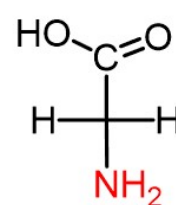
L-Glyceraldehyde



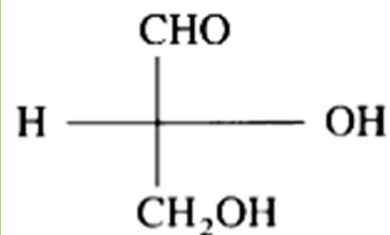
L-Alanine



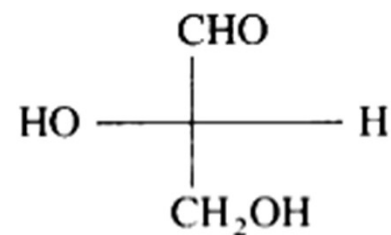
L-Serine



Glycine-achiral



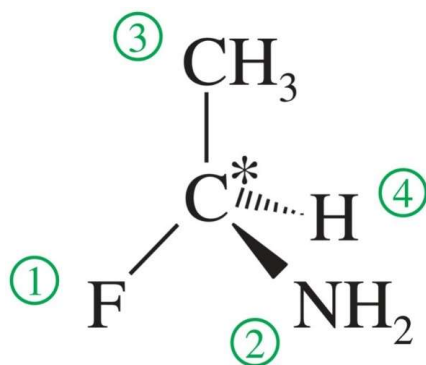
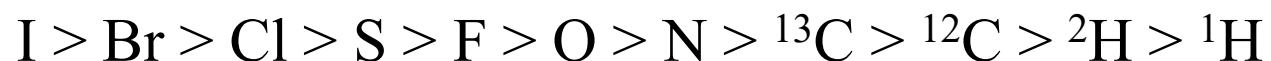
D-Glyceraldehyde
(+) Glyceraldehyde



L-Glyceraldehyde
(-) Glyceraldehyde

(*R*) and (*S*) Configuration: Step 1 **Assign Priority**

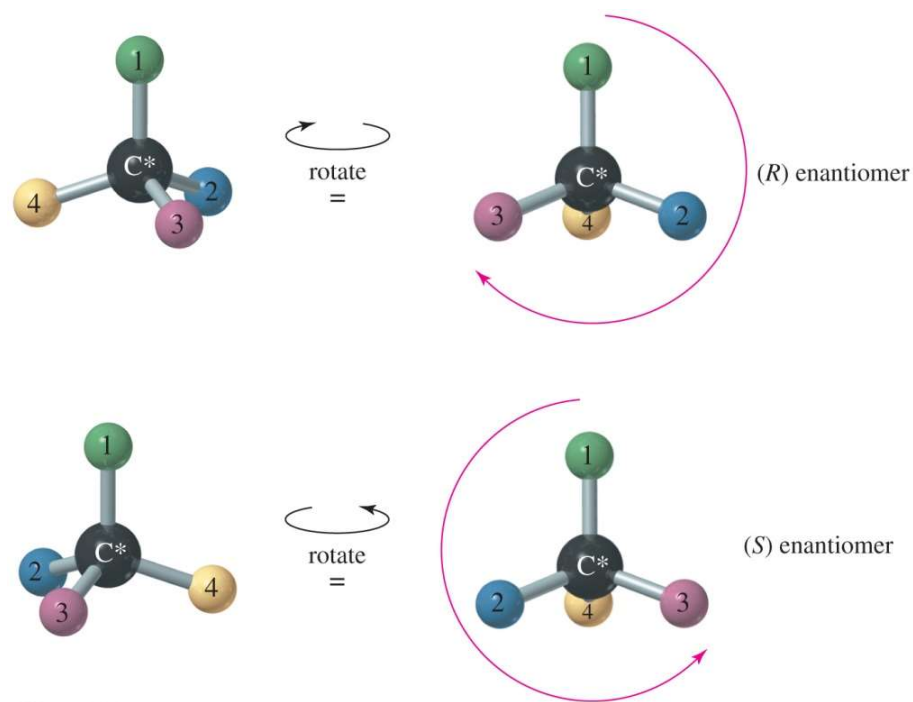
- Assign a relative “priority” to each group bonded to the asymmetric carbon.
- Atoms with higher atomic numbers receive higher priorities.

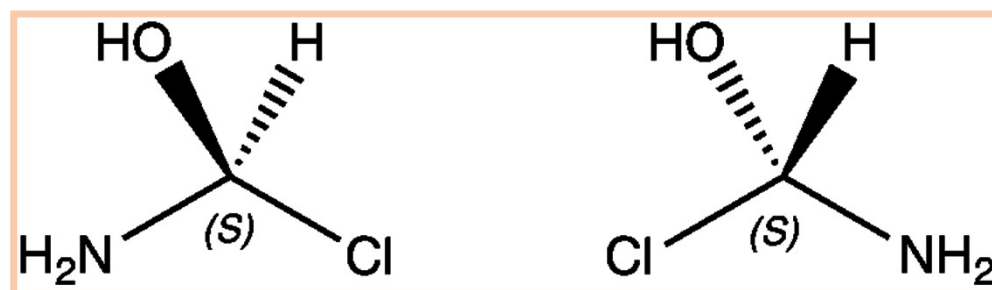
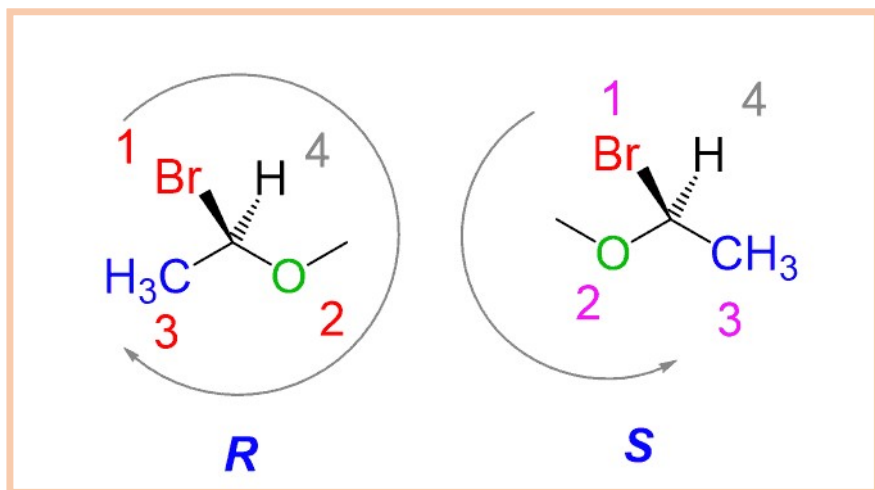
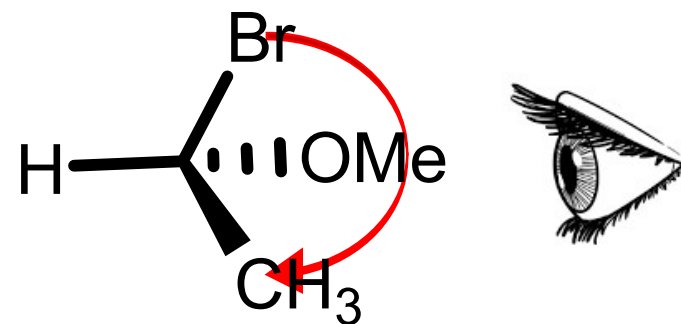
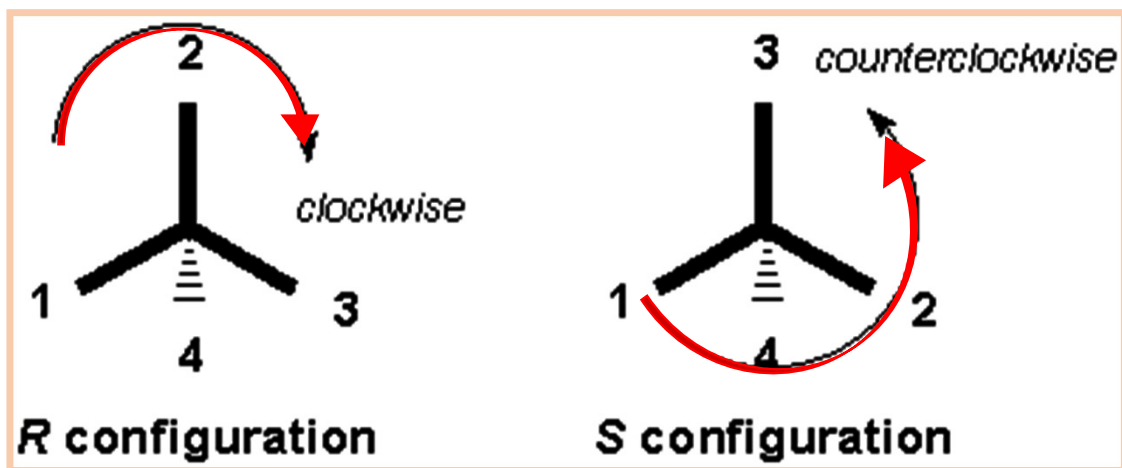


Atomic number: $\text{F} > \text{N} > \text{C} > \text{H}$

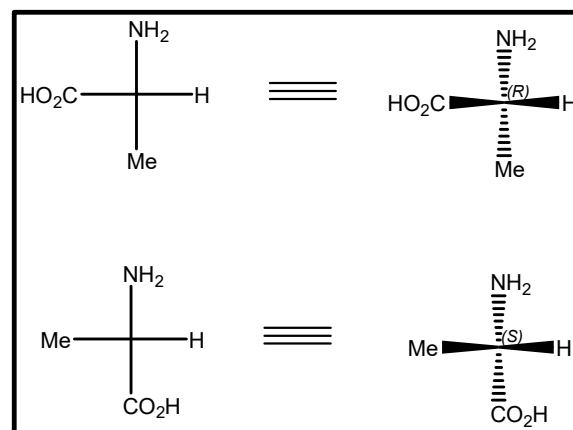
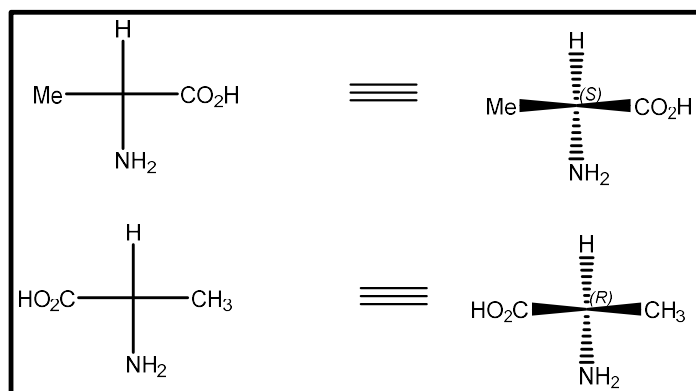
(R) and (S) Configuration: Step 2

- Working in 3-D, rotate the molecule so that the lowest priority group is in back.
- Draw an arrow from highest (1) to second highest (2) to lowest (3) priority group.
- Clockwise = (*R*), Counterclockwise = (*S*)



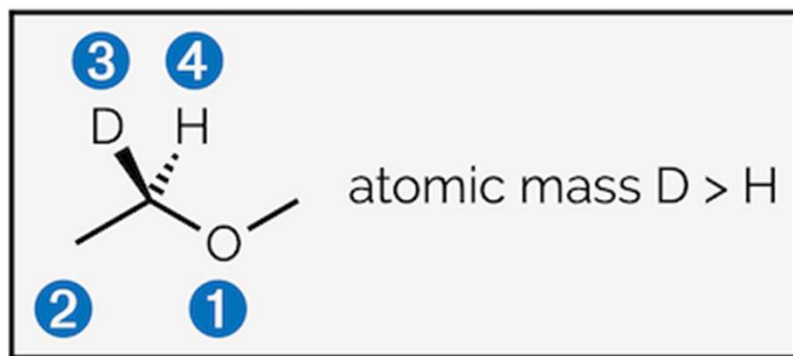


Determine R and S Configurations on a Fischer Projection

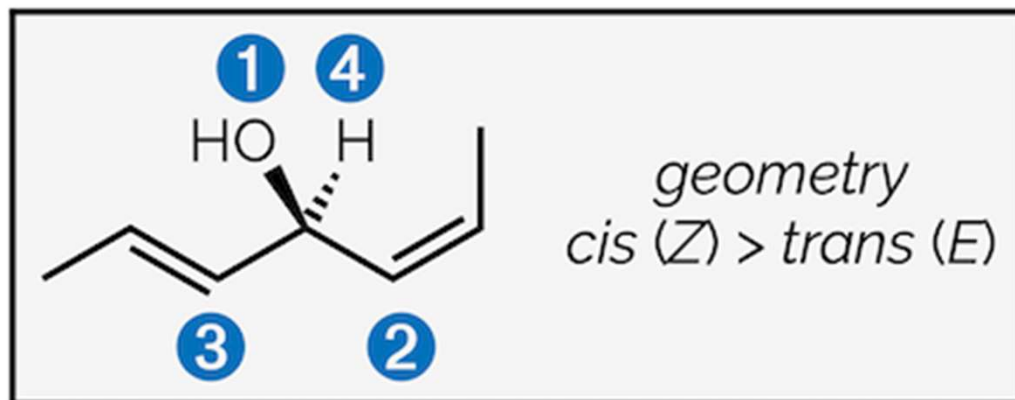
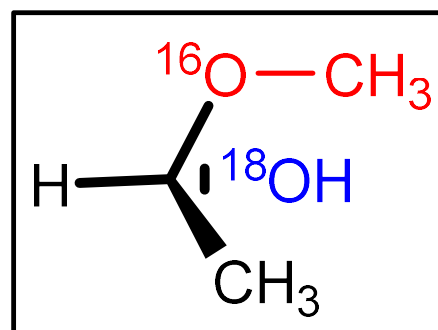


With the #4 ranked substituent at Vertical Line (pointing “in the back”), If that path is clockwise, the chiral center is *R*. If the path is counter-clockwise, the chiral center is *S*.

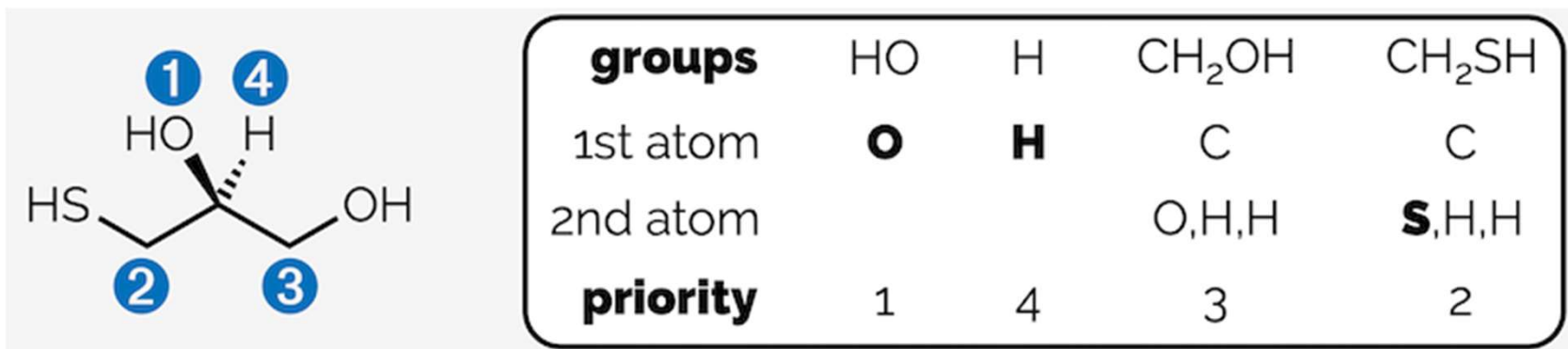
With the #4 ranked substituent at Horizontal Line (pointing “in the Front”), If that path is clockwise, the chiral center is *S*. If the path is counter-clockwise, the chiral center is *R*.



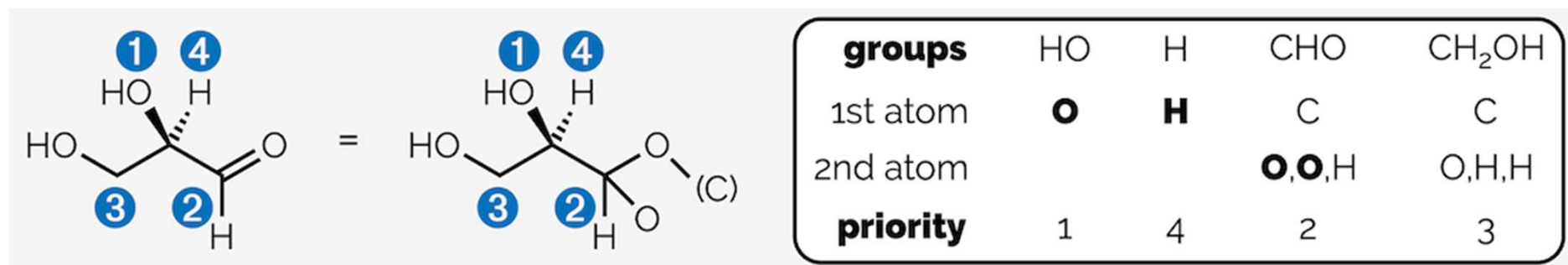
Higher atomic mass precedes
lower atomic mass (isotopes)



Cis (Z) has higher priority
than *trans (E)*



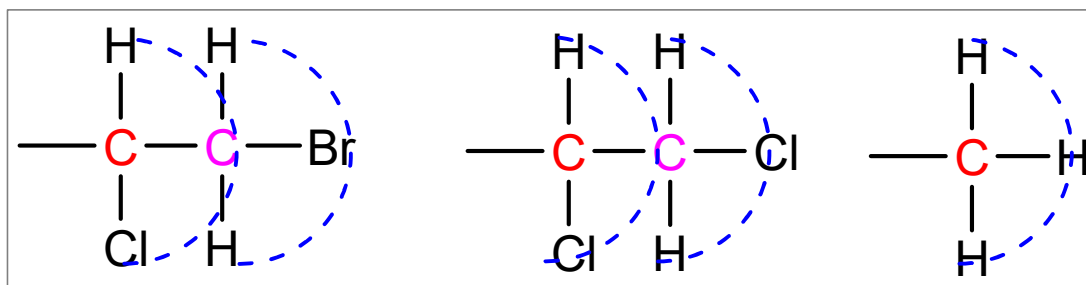
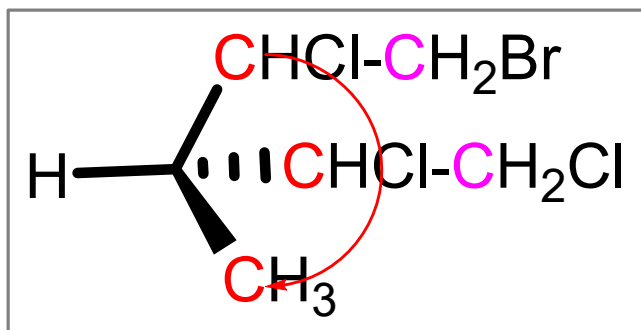
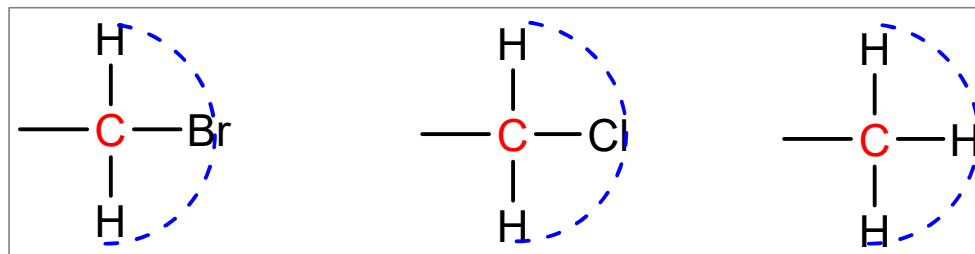
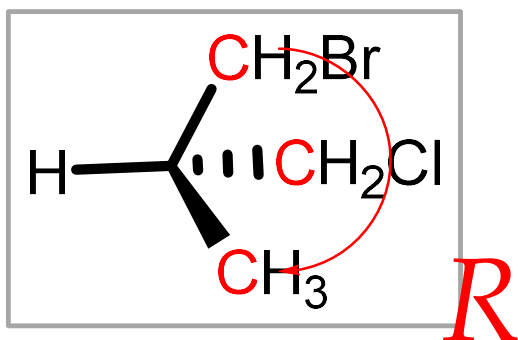
If the atoms directly attached to the chiral center are the same, then compare the next atom in the chain until a difference is found

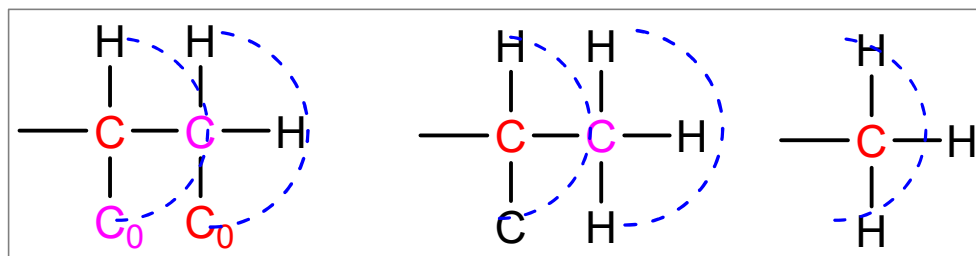
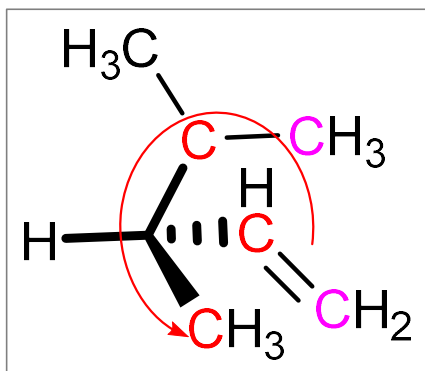


Multiple bonds have the same number of single bonds...Breaking rules

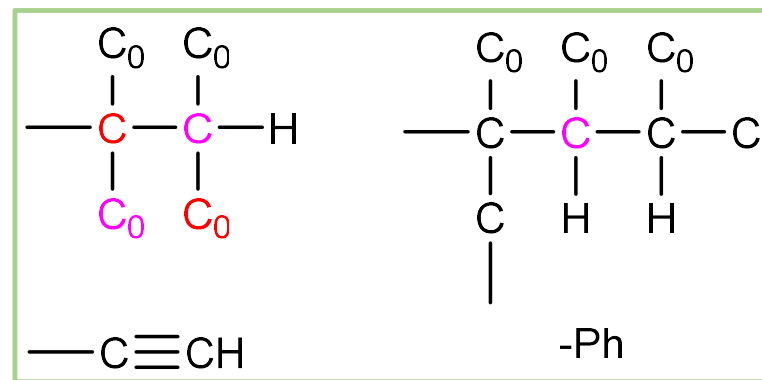
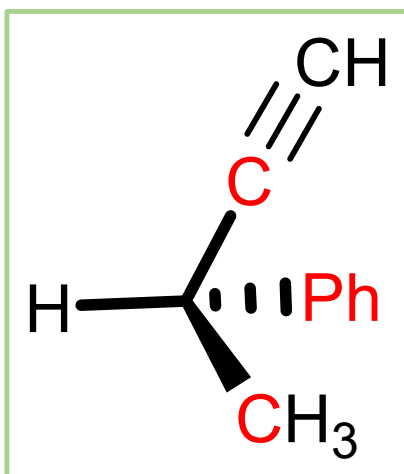
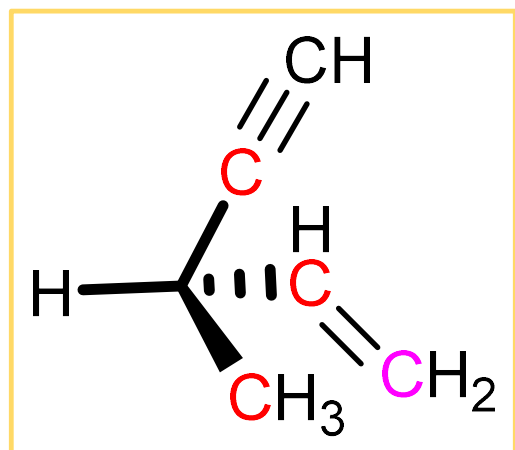
(*R*) and (*S*) Configuration: Assign Priority (Breaking Ties)

In case of ties, use the next atoms along the chain of each group as tiebreakers.

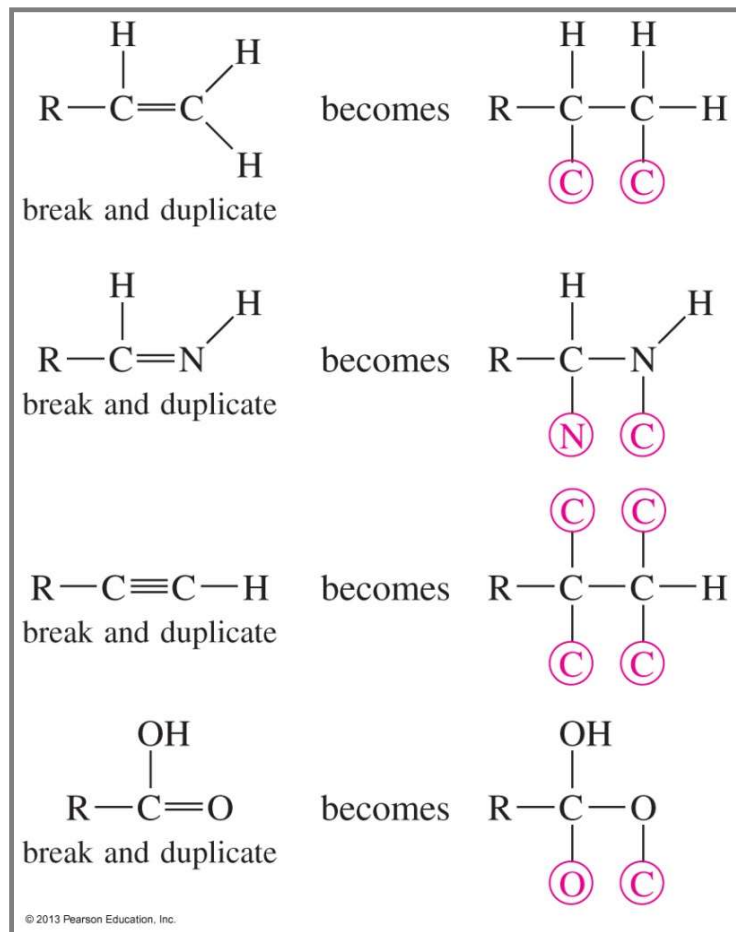




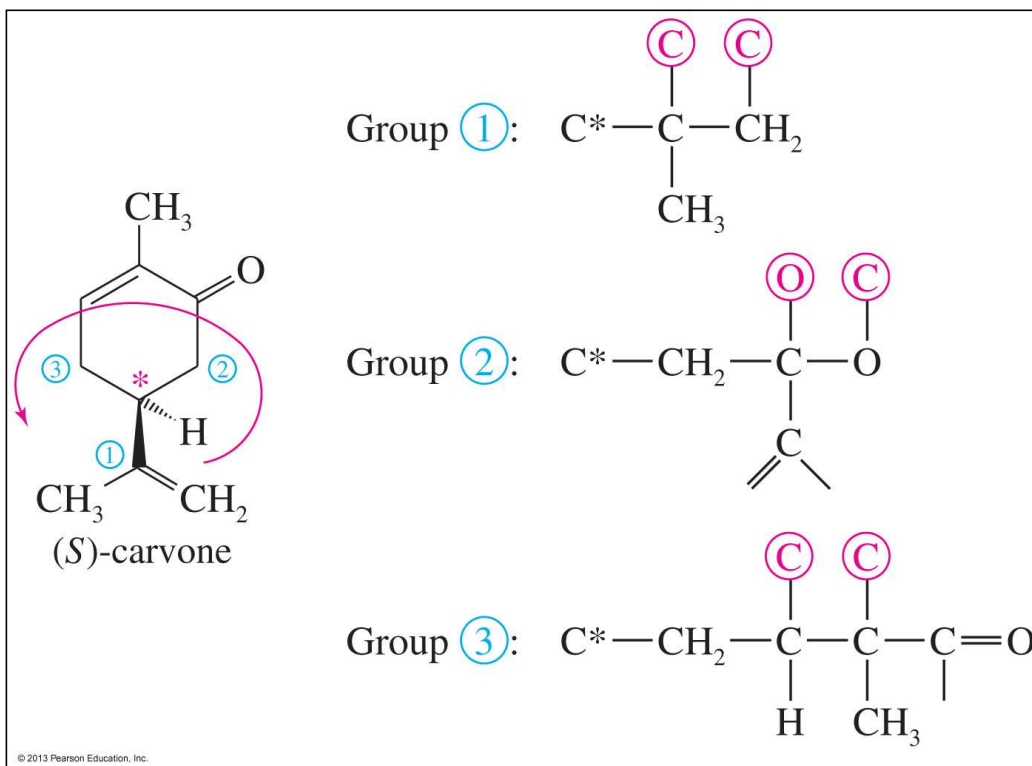
S

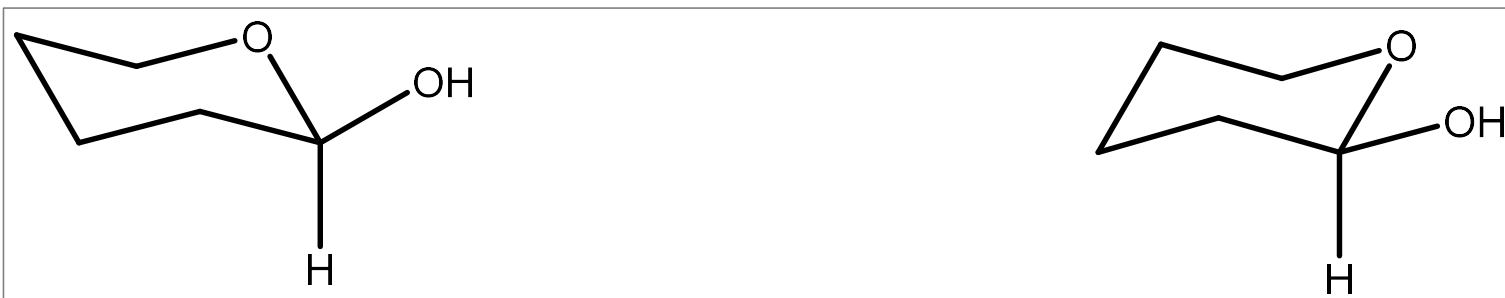


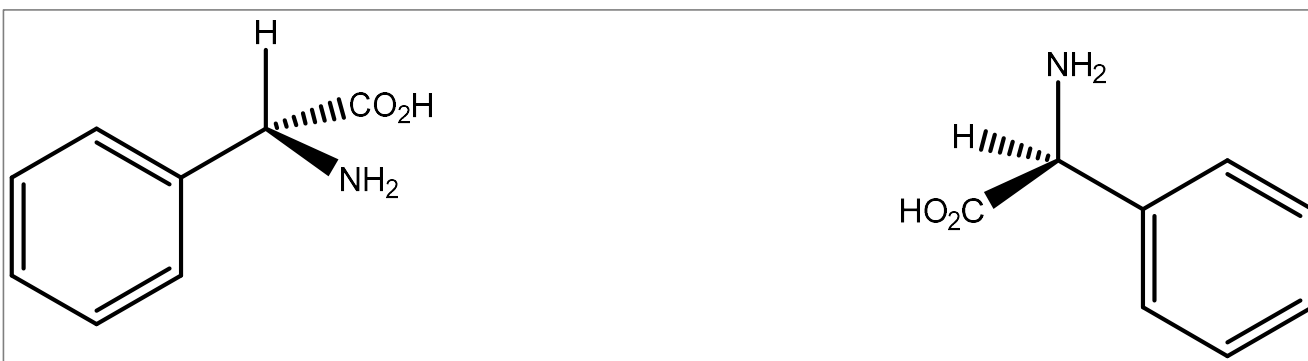
(R) and *(S)* Configuration: Step 1 **Assign Priority** (Multiple Bonds)

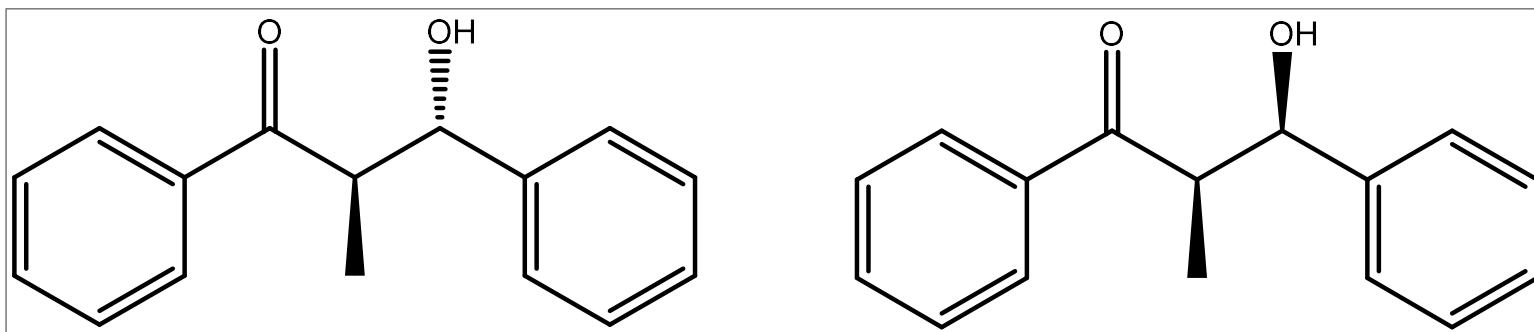


Configuration in Cyclic Compounds

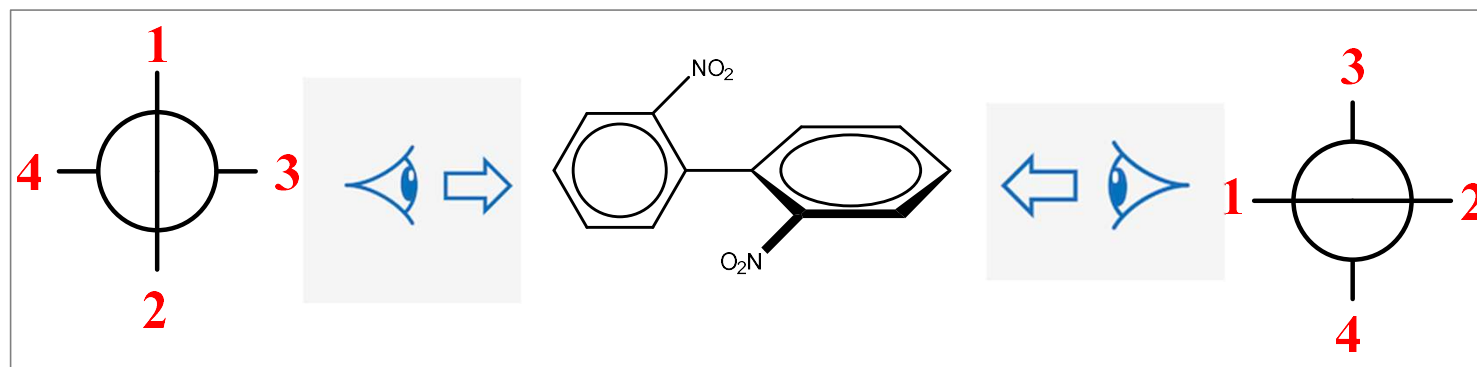
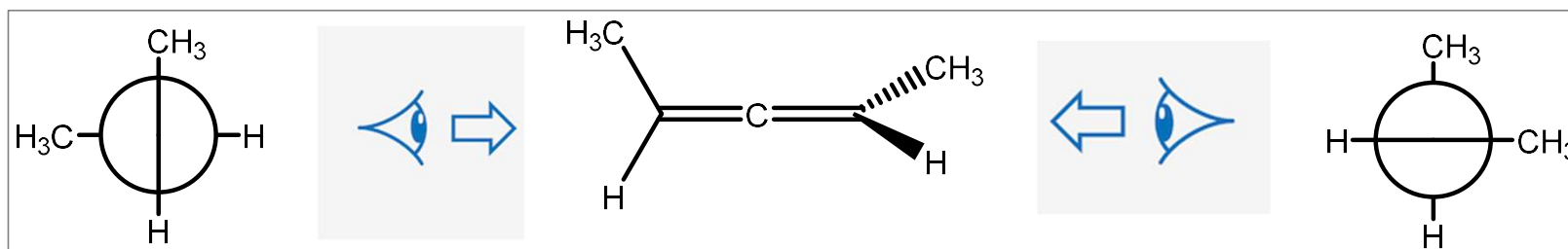








axial chirality: Nomenclature



Importance of chiral chemistry

- All proteins, enzymes, amino acids, carbohydrates, nucleosides and a number of alkaloids and hormones are chiral compounds.
- 56% of the drugs currently in use are chiral products and 88% of the last ones are marketed as racemates consisting of an equimolar mixture of two enantiomers.
- Cells are mostly composed of chiral molecules.

